



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Analyze the Impact of Noise on Image Quality and System Accuracy

ITA 0514-COMPUTER VISION

TECHNICAL PRESENTATION

Presented by : P.VIGNESH (192525013)

Department : B.Tech AI&ML



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Introduction

- Noise is an unwanted disturbance in a digital image.
- It reduces image quality and affects Computer Vision performance.
- It can occur during image capture, transmission, or storage.

Types:

- Gaussian Noise- Random brightness variations.
- Salt and Pepper Noise- Black and white dots.
- Speckle Noise- Common in radar and medical images.
- Poisson Noise- Caused by low light conditions.



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Impact on System Accuracy

Noise causes:

- Lower object detection accuracy
- Incorrect image classification
- False positives and false negatives
- Increased processing time

Workflow:

Image → Noise Added → Poor Image Quality → Wrong Detection → Reduced Accuracy



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Impact on Image Quality

- Loss of image details.
- Reduced contrast
- Poor visibility.
- Blurred edges.
- Difficult feature extraction.



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Noise Reduction Techniques

Median Filter

- * Replaces each pixel with the median value of neighboring pixels.
- * Best for Salt-and-Pepper noise.
- * Preserves edges better than the mean filter.

Gaussian Filter

- * Uses a Gaussian distribution to smooth the image.
- * Effective for Gaussian noise.
- * Slightly blurs edges.



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Thank You